

ORGANIC CHEMISTRY – CRUDE OIL & HYDROCARBONS

KEY VOCABULARY

Crude oil: a finite mixture of hydrocarbons from ancient organisms.

Hydrocarbon: a compound of hydrogen and carbon only.

Alkane: a saturated hydrocarbon, formula C_nH_{2n+2} .

Finite resource: one that will run out — not replaced quickly.

COMMON MISCONCEPTIONS

~~Crude oil is a single compound.~~

✓ It is a mixture of many hydrocarbons.

~~Hydrocarbons contain oxygen.~~

✓ They contain only hydrogen and carbon.

~~Crude oil is renewable.~~

✓ It is finite — formed over millions of years.

ALKANE FORMULAE

General formula: C_nH_{2n+2}

Methane CH_4 · Ethane C_2H_6

Propane C_3H_8 · Butane C_4H_{10}

Each carbon has single bonds only (saturated).

Longer chain → higher boiling point.

1

DEFINE HYDROCARBON

State what is meant by a **hydrocarbon**. [1 mark]

2

ALKANE FORMULA

Give the formula of the alkane with **3 carbon atoms**. [1 mark]

3

PROPERTIES & SIZE

Explain how boiling point changes as hydrocarbon chains get longer. [2 marks]

4

ORIGIN

Describe how crude oil was formed. [2 marks]

5

FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Crude oil is a ____ of hydrocarbons and is a ____ resource.

mixture

finite

compound

renewable

6

SATURATED

Explain why alkanes are described as saturated. [1 mark]